TL-Engine – Text Output (v1.09)

Simple text output is supported by the TL-Engine. The methods used are summarised in the code fragments below.

First you need to load a font ***in the game set-up area of your code***:

```
// Use the type IFont, you load new fonts in a similar way to meshes with
// a line like this:
IFont* myFont = myEngine->LoadFont( "Font1.bmp" );
```

Fonts are stored in bitmap files (.bmp). There are two supplied fonts, 'Font1.bmp' and 'Font2.bmp'.

Then you can draw text with this font, the text will only be drawn for one frame, so to keep the text permanent you must put the code ***in the game loop.*** Here is an example:

```
// This will draw text at the coordinates (200, 100) on the screen
myFont->Draw( "Hello World", 200, 100 );
```

There are optional parameters to the 'Draw' method - here is a more complex example:

```
// This will draw red text, centre aligned at the coordinates (100, 300)
myFont->Draw( "Hello Again", 100, 300, kRed, kCentre );

// Other colours are kWhite, kBlack, kRed, kGreen, kBlue, kCyan, kMagenta,
// kYellow, kLightGrey & kDarkGrey, or you can use RGB hexadecimal integers.
```

To output more complex text (including variables or functions/methods) use the 'stringstream' type, which can be used like 'cout' to create a string:

```
// Need to include the sstream library for stringstream support. Add these lines
at the top of the program
#include <sstream>
using namespace std;

// ...
// ...

// This code will output the X & Z coordinates of a white ball to the screen
stringstream outText;
outText << "White Position: " << whiteBall->GetX() << " " << whiteBall->GetZ();
myFont ->Draw( outText.str(), 20, 20 );
```

Note: Outputting newlines (endl or '\n') has no effect in the TL-Engine - use more than one 'Draw' call instead.